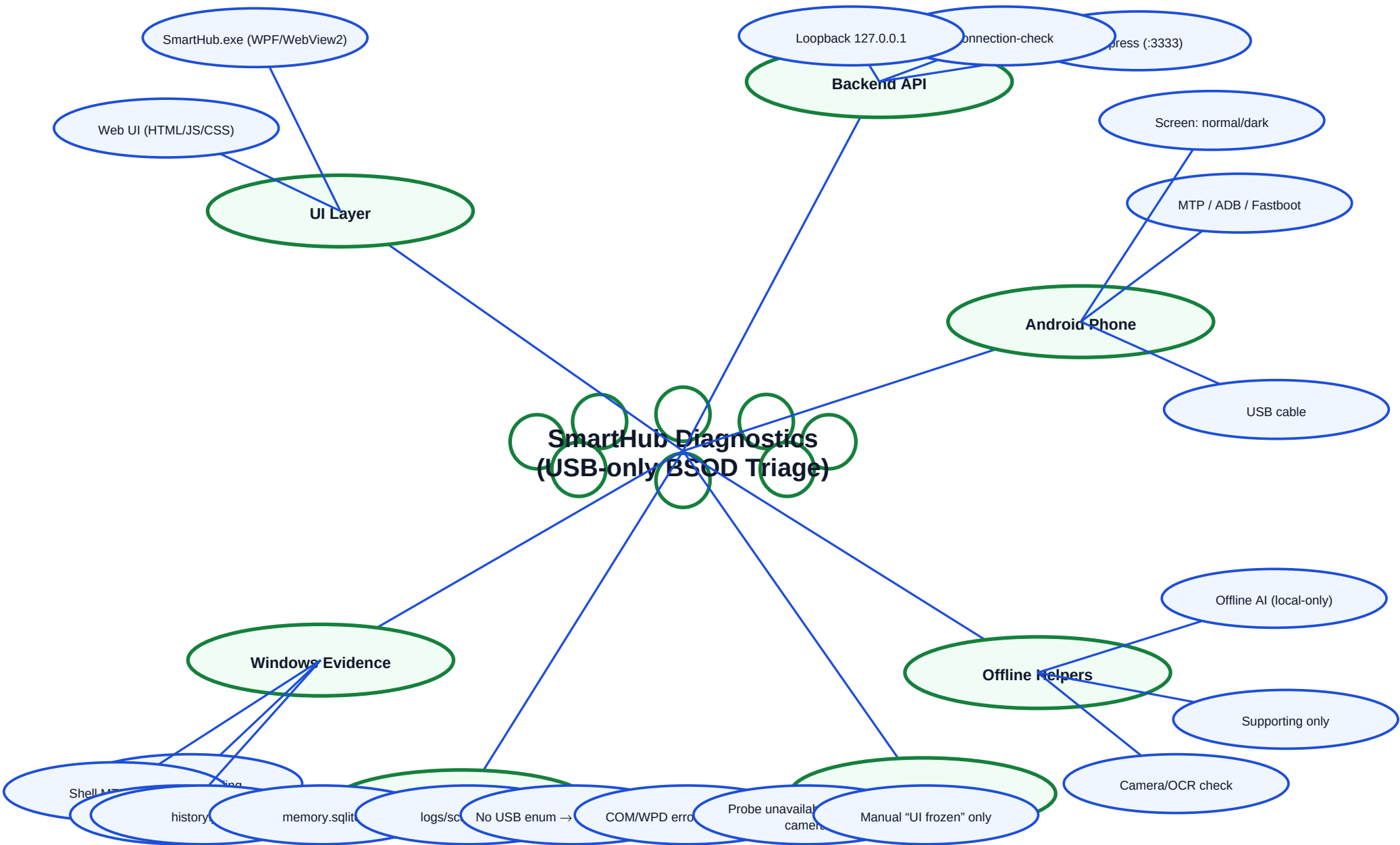


SmartHub Diagnostics – Conceptual Mapping

Generated: 2026-04-12 | Scope: USB-only BSOD triage + offline helpers



Conceptual Mapping (Narrative)

- 1) Technician opens SmartHub (WPF/WebView2). The embedded Web UI runs locally.
- 2) Web UI calls the local Node backend (127.0.0.1:3333), mainly /connection-check for USB-only triage.
- 3) Backend collects Windows evidence (PnP/CIM USB sampling, optional WPD helper, Shell-based MTP probe fallback).
- 4) Optional camera/OCR and offline AI are best-effort supporting tools; they must not override truthfulness rules.
- 5) Safety rule: if the PC cannot enumerate the phone over USB, the result is INCONCLUSIVE.
- 6) Host limitation rule: if WPD/COM is broken (E_NOINTERFACE), the app outputs host-inconclusive guidance and does not infer the phone's state. Webcam hints are ignored; only manual "UI frozen" confirmation can record freeze.

Shape meanings

- Cloud (center) = the main concept/system being mapped.
- Large ovals = major components/actors (primary branches).
- Small ovals = subcomponents, signals, or rules (supporting details).
- Lines = conceptual relationships / information flow (not strict sequencing).

Connection Relations (Text Mode)

These lines summarize how parts connect (A → B means A calls/uses/sends data to B).

UI ↔ Backend

- SmartHub.exe (WPF + WebView2) → Web UI (HTML/JS/CSS): hosts/embeds the UI.
- Web UI → Node Backend (Express :3333): loopback HTTP requests (127.0.0.1).
- Node Backend → Web UI: JSON response used to render diagnostic status/result.

USB Evidence

- Node Backend → Windows Host OS: PowerShell/CIM/PnP enumeration + USB stability sampling.
- Node Backend → Shell MTP probe: best-effort check for MTP accessibility without WPD COM dependency.
- Node Backend → UsbEvidenceHelper (WPD/COM, optional): deeper MTP/WPD probing when available.
- Windows Host OS ↔ Android Phone: USB cable enumeration (MTP/ADB/Fastboot interfaces when present).

Offline Helpers

- Node Backend → Camera/OCR helper: optional visual hints (screen dialog / darkness) when supported.
- Node Backend → Offline AI helper: optional local-only suggestion based on collected signals.

Storage

- Node Backend → Local artifacts: writes logs, run history, and (optional) screenshots/AI cases.
- Web UI → Local storage/history view: reads prior runs for technician review (implementation-specific).

Truthfulness Gates (Always Apply)

- No Windows USB enumeration → output INCONCLUSIVE (do not claim phone BSOD/freeze).
- Host COM/WPD error → host-inconclusive (do not infer phone state; ignore camera hints).
- Manual “UI frozen” checkbox → only way to record freeze when probing is unavailable.